

Venkata Sai Aravind Devarampati

Embedded Systems Engineer | C | RTOS & Bare-Metal Development
IoT | Firmware | Secure Systems

📍 Blacksburg, VA ✉ venkatsaiaravind@vt.edu 📞 +1-(540)-757-3557

in <https://linkedin.com/in/aravind-devarampati>

Professional Summary

An Embedded Systems Engineer specializing in firmware for ARM Cortex-M, spanning bare-metal and FreeRTOS systems. Experienced in real-time scheduling and low-level driver development (SPI/I2C/UART). Research background in secure embedded systems and electromagnetic fault injection (EMFI), with strong debugging skills using GDB, oscilloscopes, and logic analyzers.

Technical Skills

Programming: C/C++, Python, Embedded C, Kernel Programming, Bash, Verilog.

Architectures: ARM Cortex-M (NVIC, MPU, TrustZone), ARM Cortex-A.

Hardware Debugging & Tools: GDB, SWD, Oscilloscopes, Logic Analyzers, UART Debuggers.

Protocols & Interfaces: UART, SPI, I²C, USB, BLE, Wi-Fi, LoRaWAN.

RTOS & Embedded OS: FreeRTOS, Embedded Linux (Device Drivers, Kernel Modules), Ubuntu, Raspbian OS.

Hardware: STM32, MSP432 LaunchPad, Raspberry Pi, Arduino Mega 2560, Riscure Tools.

Dev Ecosystem: Makefiles, Linker Scripts, STM32CubeIDE, KEIL/IAR, OpenOCD, Git.

Research Experience

Research Assistant, BEARHW Lab

May 2026 - Present

DRAM Retention Characterization for Hardware-Assisted Oblivious Transfer on FPGA SoCs

- Developed bare-metal C firmware on a LiteX-based RISC-V SoC on Xilinx ZCU104 to support hardware-assisted evaluation of Oblivious Transfer workflows and memory-state experiments.
- Designed and automated DRAM-retention testing by programming deterministic patterns, targeting specific DIMM regions, running controlled delay schedules, and capturing memory snapshots for offline analysis.
- Investigated and modified LiteDRAM refresh-control behavior, memory timing parameters, and retention-run orchestration to create a reproducible experimental platform for long-duration decay analysis.

Research Assistant, BEARHW Lab

Sept 2024 - May 2025

Electromagnetic Fault Injections

- Conducted Electromagnetic Fault Injection (EMFI) using the Riscure platform, performing systematic parameter sweeps across timing, voltage, and spatial coordinates to compromise secure targets.
- Exploited the AES-128 engine on the Riscure Piñata board by inducing and characterizing hardware faults to identify critical execution windows vulnerable to key leakage.
- Engineered a fault-resilient firmware stack featuring instruction-level countermeasures, secure exception handling, and tamper-response logic to mitigate the effects of adversarial EM pulses.
- Quantified reliability of secure boot and encryption routines under repeated fault injection and developed a threat model to guide future tamper-resistant hardware-software interfaces.

Teaching Experience

Graduate Teaching Assistant

Jan 2025 - May 2025

Virginia Tech

- Assisted Dr. Leyla Nazhandali in teaching ECE 2564: Embedded Systems, covering embedded C, microcontroller programming, and hardware-software interfacing.
- Designed a reference interrupt-driven FSM framework in bare-metal C on MSP432 and guided students in extending it with non-blocking real-time logic, interrupt handling, and peripheral integration.

Projects

Tier-aware Allocator for Heterogeneous Memory Systems

Mar 2025 - May 2025

Linux Kernel, C, NUMA, CXL, PMEM, QEMU/KVM

- Architected a tier-aware allocator for DRAM/CXL/NVM systems; optimized data placement by quantifying an $8.5\times$ CXL latency penalty and $40\times$ NVM throughput gap to align workload sensitivity with hardware performance.

Energy-Aware Task Scheduling with Runtime Frequency Scaling

Mar 2025 - Apr 2025

STM32F446RE, ARM Cortex-M4, PLL, Register-level C

- Implemented runtime multi-level CPU frequency scaling on STM32F446RE using PLL reconfiguration to reduce energy consumption under mixed real-time workloads.

Real-Time Interrupt Latency Optimization on STM32

Feb 2025 - Apr 2025

FreeRTOS, ARM Cortex-M, NVIC, Logic Analyzers

- Measured and optimized interrupt latency and jitter on ARM Cortex-M by tuning NVIC priorities, enabling interrupt nesting, and deferring long ISR work to tasks to improve real-time determinism under FreeRTOS.

Memory Reclamation for Lock-Free Concurrent Data Structures

Oct 2024 - Dec 2024

C++, Atomics, Multithreading

- Developed Hazard Pointer and EBR schemes for lock-free structures; eliminated ABA/use-after-free hazards and improved scalability across 40+ threads with quantified latency advantages.

Secure Execution Environment on ARM Cortex-M

Mar 2024 - Apr 2024

STM32, ARMv7-M, MPU, C, Assembly

- Designed a Trusted Execution Environment on STM32 using MPU-based memory isolation, privilege separation, and secure interrupt handling.

Education

Virginia Tech

M.S. in Computer Engineering

GPA: 3.96/4.0

Dayananda Sagar College of Engineering

B.E. in Electronics & Communication Engineering

GPA: 8.53/10.0

Certifications

- ARM Cortex-M Architecture & Software Development Specialization
- Architecting Smart IoT Devices (EIT Digital)